1.2 The Principle of Least Action in Classical Mechanics

In Section 1.1, we employed the calculus of variations—that is, we considered infinitesimal variations of a function and used their commutativity with differentiation to derive the governing equation—to obtain the solution to the brachistochrone problem. The mathematical essence of this method is to establish a relationship between infinitesimal changes in a function and the corresponding variations of a functional. In the brachistochrone problem, the functional was constructed from the total travel time of the motion.

In classical mechanics, if the position x and velocity v of a particle are known, its subsequent motion can be determined. In field theory, we regard the world as being composed of different superimposed fields. Under the influence of these fields, represented by the action S, a particle moves through the world according to the principle of stationary action. We begin by writing down the expression for the action:

$$S = \int_{t_1}^{t_2} L(x, v)dt \tag{1.2.1}$$

Following the same procedure as in Eq. (1.1.7), we take the variation of the action and obtain

$$\begin{aligned}
\delta S = \int_{x_1,t_1}^{x_2,t_2} \left(\frac{\partial L}{\partial x}\delta x + \frac{\partial L}{\partial v}\delta v\right) dt &= \int_{x_1,t_1}^{x_2,t_2} \left(\frac{\partial L}{\partial x}\delta x + \frac{\partial L}{\partial v}\delta \frac{dx}{dt}\right) dt \\
&= \int_{x_1,t_1}^{x_2,t_2} \frac{\partial L}{\partial x}\delta x dt + \int_{x_1,t_1}^{x_2,t_2} \frac{\partial L}{\partial v} d\delta x \\
&= \int_{x_1,t_1}^{x_2,t_2} \frac{\partial L}{\partial x}\delta x dt + \frac{\partial L}{\partial v}\delta x\Big|_{x_1,t_1}^{x_2,t_2} - \int_{x_1,t_1}^{x_2,t_2} \delta x d\left(\frac{\partial L}{\partial v}\right) \\
&= \int_{x_1,t_1}^{x_2,t_2} \left[\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial v}\right)\right] \delta x dt
\end{aligned} \tag{1.2.2}$$

We thus obtain the Euler–Lagrange equation for classical mechanics:

$$\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial v}\right) = 0 \tag{1.2.3}$$

This is the general equation governing the motion of a particle. Here, L is called

the **Lagrangian**, which is a function of the particle's position, velocity, and time, namely $L(x, v, t)$. We now examine the form and some properties of the Lagrangian in classical mechanics.

According to the homogeneity of time and space, changing the origins of the time and spatial coordinates does not alter the form of L. Therefore, the Lagrangian cannot depend explicitly on the position x or time t. Furthermore, owing to the isotropy of space, the Lagrangian should depend only on the squared magnitude of the velocity, v^2. Under these conditions, Eq. (1.2.3) becomes

$$\frac{d}{dt}\left[\frac{\partial L(v^2)}{\partial v}\right] = \frac{d}{dt}\left[\frac{\partial L(v^2)}{\partial v^2}\cdot\frac{\partial v^2}{\partial v}\right] = \frac{d}{dt}\left[\frac{\partial L(v^2)}{\partial v^2}\cdot 2v\right] = 0 \tag{1.2.4}$$

Equation (1.2.4) indicates that the expression inside the square brackets must be a constant. If v is not constant, then $\partial L(v^2)/\partial v^2$ would have to be proportional to the reciprocal of v, which is inconsistent with the conclusion that L is a function of v^2. Therefore, Eq. (1.2.4) leads to two conclusions: the velocity of a freely moving particle remains constant, and

$$L = kv^2 \tag{1.2.5}$$

For a free particle, we conventionally set $k = m/2$, so that the Lagrangian takes the form of the kinetic energy. In addition, using Eq. (1.1.16), we obtain

$$\frac{\partial L}{\partial v}\cdot v - L = C \tag{1.2.6}$$

For a conservative system, the quantity conserved in time is the total energy of the system. Thus,

$$\varepsilon = \varkappa = \frac{\partial L}{\partial v}\cdot v - L \tag{1.2.7}$$

In classical mechanics, the total energy consists primarily of two contributions: kinetic energy and potential energy. According to Eq. (1.2.5), we have

$$\frac{\partial L}{\partial v}\cdot v = mv^2 = 2T \tag{1.2.8}$$

Substituting Eq. (1.2.8) into Eq. (1.2.7), we can determine the form of the Lagrangian:

$$L = 2T - \varkappa = T - U \tag{1.2.9}$$

Here, T denotes the kinetic energy and U denotes the potential energy.

Substituting Eq. (1.2.9) into Eq. (1.2.3), we obtain Newton's second law:

$$m\frac{dv}{dt} = -\frac{\partial U}{\partial x} \tag{1.2.10}$$